

Nmap

CompTIA PenTest+ (PT0-003)

Domains:

- 2.0: Reconnaissance and Enumeration
- 3.0: Vulnerability Discovery and Analysis

Objectives:

- 2.2: Given a scenario, apply enumeration techniques
- 3.1: Given a scenario, conduct vulnerability discovery using various techniques.

Video Overview

This video introduces Nmap, a fundamental tool for network discovery and vulnerability scanning. As you embark on or advance in a career in cybersecurity, understanding tools like Nmap is essential. This video will demonstrate how Nmap helps reveal hidden hosts, services, and vulnerabilities within a network. The concepts discussed are directly applicable to real-world penetration testing, helping you to identify and address security weaknesses in various systems and environments.

Key Topics Covered

- Nmap's role in network discovery and vulnerability scanning
- Common Nmap scan types: Stealth Scan (-sS), TCP Connect Scan (-sT), and UDP Scan (-sU)
- Host discovery and the -PN option
- Timing options (-T0 through -T5) for controlling scan speed
- Service Version Detection (-sV) for identifying software versions
- OS Detection (-O) for identifying target operating systems
- Nmap Scripting Engine (NSE) and the --script argument
- Aggressive Scan (-A) for combining multiple functionalities
- Port specification (-p) for targeted scanning

Student Notes Section